

Spencer Lukehart

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Senior full-stack software engineer who ships end-to-end products on a **TypeScript / JavaScript / Nuxt / Node.js / Postgres** stack — from data model and performant APIs to polished Vue/Nuxt front-ends — and builds production LLM integration and agentic features into them. High agency in ambiguous problem spaces: turns design into running code in days, not weeks, working directly with product stakeholders.

SKILLS

- **AI / LLM & Agents:** LLM integration, agentic systems & agent development, evaluation frameworks (evals), MCP (Model Context Protocol) servers & connectors, retrieval systems — code knowledge graph (FalkorDB), multi-channel agent actions (Slack / Jira / GitHub), scheduled autonomous agents (Claude Code routines), tool use, advanced prompt engineering, AI coding tools (Claude Code, Cursor, ChatGPT)
- **Backend & APIs:** Node.js, Express.js, TypeScript, JavaScript (ES6+), Python, Java, RESTful API design, microservices, async/event-driven systems, transaction management, structured logging, standardized error responses
- **Frontend:** modern JavaScript frameworks — Vue.js (2 & 3), Nuxt.js (Nuxt 3 / Nitro), SSR (Server-Side Rendering), SPA architecture, single-spa micro-frontends, Composition API, Pinia, Vite, Webpack, HTML5/CSS3
- **Cloud, Infra & DevOps:** AWS (Lambda, S3, EC2, ECS, CloudWatch), GCP (Cloud Functions), serverless architecture, Docker, GitHub Actions, CircleCI, CI/CD pipeline design, Datadog (APM, RUM, Logs), observability instrumentation, Linux
- **Data:** PostgreSQL, MySQL (Amazon Aurora / RDS), Redis caching, Firestore, Snowflake, SQL performance tuning & query optimization, relational & NoSQL data modeling
- **Practices:** Test-driven development (Jest, Vitest), unit & integration testing, system design, performance optimization, code review, mentoring, Agile/Scrum, cross-functional collaboration

EXPERIENCE

Senior Software Engineer — Lasso Inc., Charleston, SC

April 2026 – Current

- Designed and shipped a production **LLM agent** that detects 500-class errors in **Datadog**, reasons over root cause, and acts across **Slack, Jira, and GitHub** — auto-fixing behind a **three-stage evaluation framework** (TDD reproduction, blast-radius / caller analysis, UI response-path evals) or escalating to the area-expert engineer when agent gates do not pass. Cut server errors **81%** as traffic grew 20%. Wired Atlassian, GitHub, and Datadog **MCP connectors** into the agent's tool layer.
- Built a **code knowledge graph** (FalkorDB) that indexes ~3,400 files, 449 routes, and 260+ data models across two repos into a queryable graph, giving AI agents **structural memory** for blast-radius and cross-repo impact analysis over **MCP**.
- Authored the team's **end-to-end observability roadmap** — correlating **database queries, API request flows, and UI sessions** by request-ID with custom APM spans — and a **scheduled autonomous agent fleet** deployed via Claude Code routines (runs every 2h; bug detection live, with slow-query, error-spike, Pendo drop-off, and frontend-performance agents next).
- Led architecture and production rollout of a **single-spa micro-frontend** running Vue 3 alongside the legacy Vue 2 app, authored the team's migration runbook, mentored engineers on Vue 3 / Composition API patterns, and led team-wide adoption of **Claude Code, Cursor, and AI-assisted development workflows** so engineers ship pages and features independently.
- As part of the Vue 3 migration, replaced **BootstrapVue** (Vue 2-only) with an **AI-native design system** on the path to **generative UI (GenUI)**: **DTCG design tokens** compiled via **Style Dictionary** into a runtime theme, headless **reka-ui + Tailwind** components, and a **machine-readable component registry** (JSON schemas for 40+ components) that AI agents consume over **MCP** — so agent-generated screens conform to the design language.
- Engineered a **cross-framework shared-state bridge** between Vue 3 and the existing Pinia stores and stood up the **parallel Vite build pipeline** (hot reload in dev, UMD bundle in prod) proxied through the existing Webpack dev server.

Software Engineer III — Lasso Inc., Charleston, SC

April 2024 – April 2026

- Built a compliance-driven warehouse management system supporting complex logistics workflows in **Node.js, Express, Vue, and Amazon Aurora MySQL**, balancing user configurability with strict regulatory requirements.
- Reduced complex function execution by **90%** by architecting a **Redis** caching layer, significantly improving responsiveness and lowering infrastructure load.
- Designed and optimized high-performance SQL queries in Amazon Aurora MySQL, maintaining SLA compliance for real-time tracking and scanning operations.
- Partnered with principal engineers on large-scale system rebuilds; led **technical discovery and end-to-end feature delivery** with product stakeholders and consistently shipped on schedule.
- Refactored legacy systems to current engineering standards, improving maintainability and team velocity.

Software Engineer II — Bear Cognition, Charleston, SC

October 2023 – April 2024

- Led migration of a legacy SPA from deprecated Nuxt 2 to Nuxt 3 with the Nitro server runtime, replacing the Google Firebase Functions backend with Nuxt server routes and decreasing application load time.
- Stood up the team's **CI/CD pipeline in GitHub Actions** — automated lint, type-check, test, and deploy jobs for every PR — replacing manual deploys.
- Architected the transition from a PostgreSQL relational database to Google Firestore for more efficient querying and stored-procedure equivalents.
- Mentored junior developers through code reviews, improving code quality and team velocity.

Software Engineer I — Bear Cognition, Charleston, SC

July 2022 – October 2023

- Implemented backend features in Node.js and Google Cloud Functions for data fetching and transformation within a Nuxt frontend.
- Engineered data pipelines integrating Snowflake, PostgreSQL, and Tableau to improve customer reporting.
- Implemented microservices via AWS Lambda to streamline internal task automation.

Developer — Zuora, College of Charleston, Charleston, SC

January 2022 – April 2022

- Built event-driven REST APIs for asynchronous synchronization between the company database and Salesforce, using Apex to transfer JSON payloads between systems.
- Worked on a four-person Scrum team with rotating Scrum Master duties.

EDUCATION

B.A. in Computing in the Arts: Digital Media — College of Charleston, May 2022 Minor in German Language Studies · Dean's List

- Member, German Honors Society

CERTIFICATIONS

- Google Cloud Platform: Digital Cloud Leader — November 2022
- Amazon Web Services: Certified Cloud Practitioner — August 2022

PROJECTS

Agentic Automated Content Generation — End-to-end news-to-video pipeline using the **Grok API** for story selection and script generation and the HeyGen API for AI-avatar video. Built in Go and Python.